1. a. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.
b. Create a PWA that displays a custom offline page when the user is not connected to the internet.

Step 1: Create a Project: Open Visual Studio 2026 → Click **Create a new project** → Select **ASP.NET Core Empty** → Name the project **CollegePortal** → Click **Create**.

Step 2: Create wwwroot

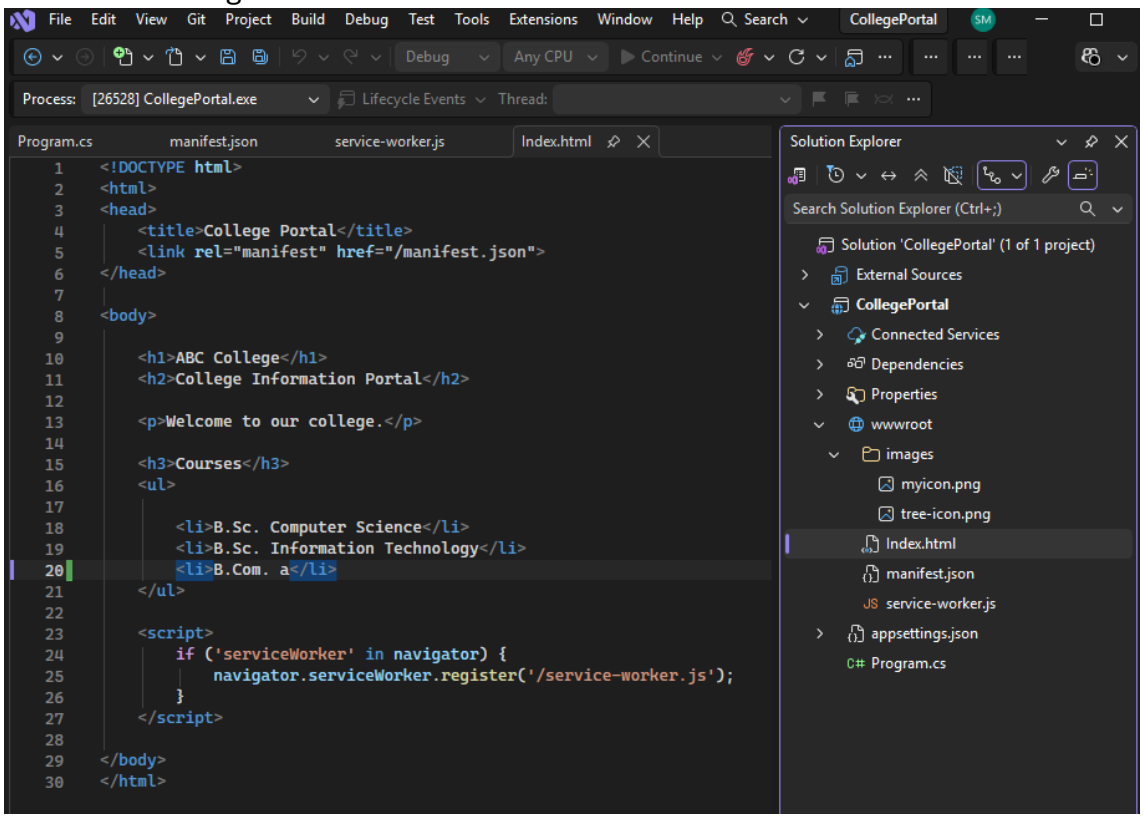
In **Solution Explorer**, right-click the project → **Add → New Folder** → name it: wwwroot
Create these three files inside wwwroot:

- index.html
- manifest.json
- service-worker.js

Also create a folder named images and place an icon named icon.png inside it.

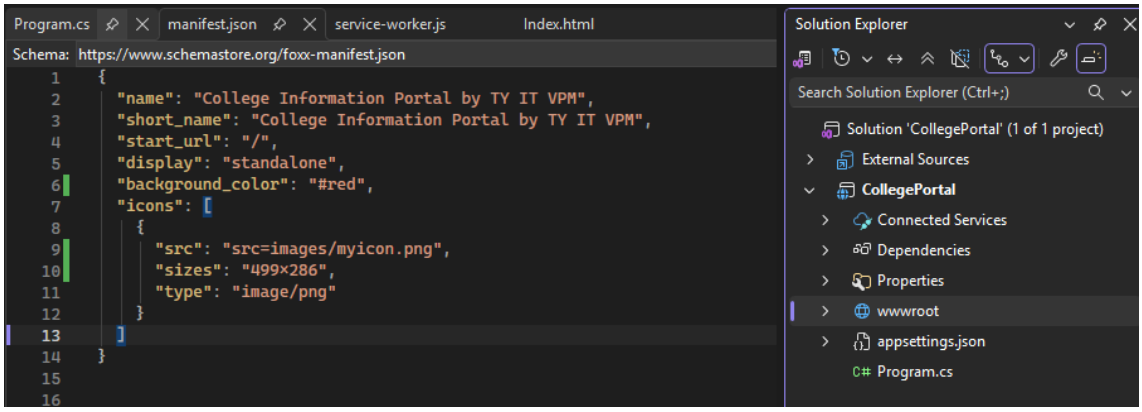
Step 3: Create the Website

Add the following code to index.html:

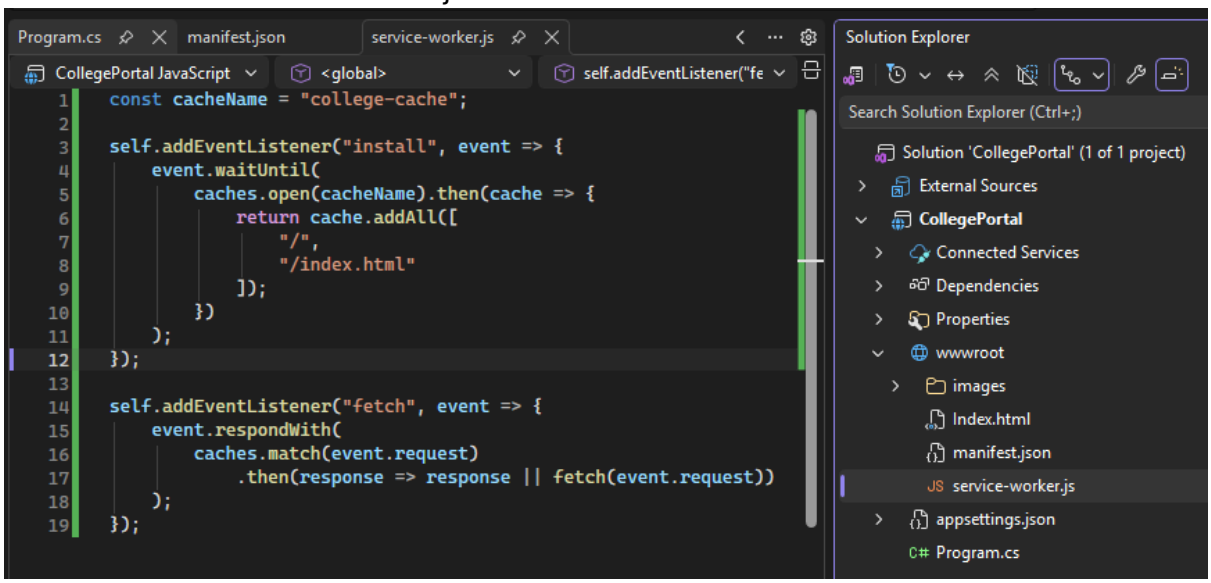


Step 4: Create the Web App Manifest

Add this code to manifest.json:



Step 5: Create the Service Worker
Add this code to service-worker.js:



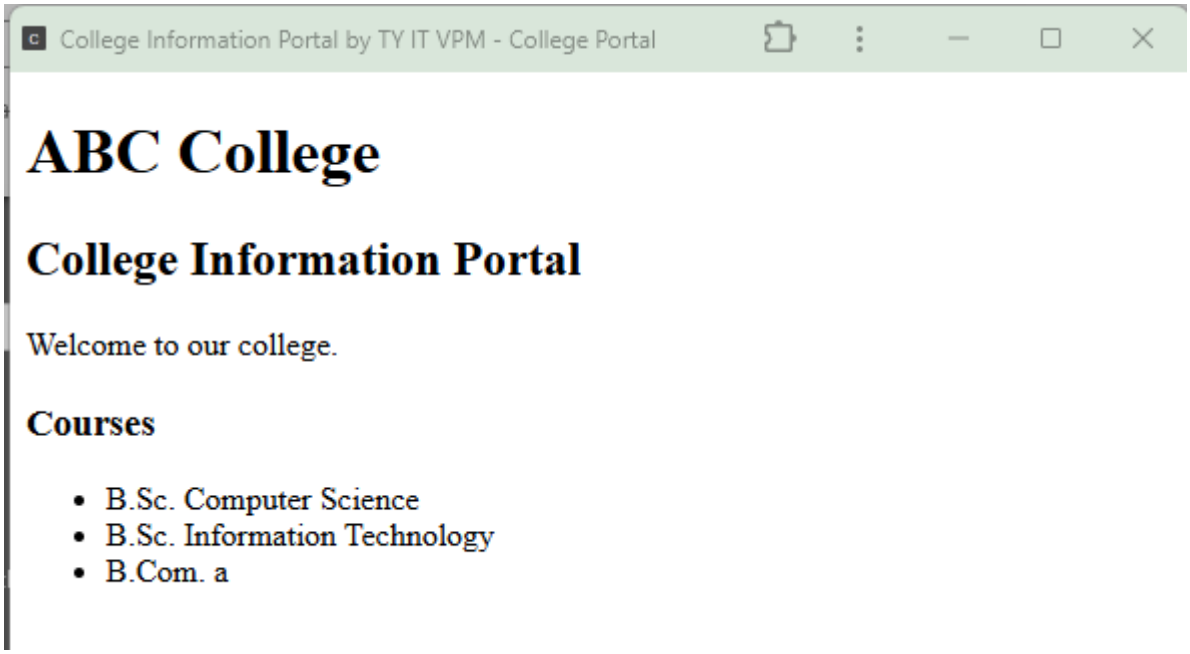
This stores the website in the **browser cache**, allowing it to work offline.

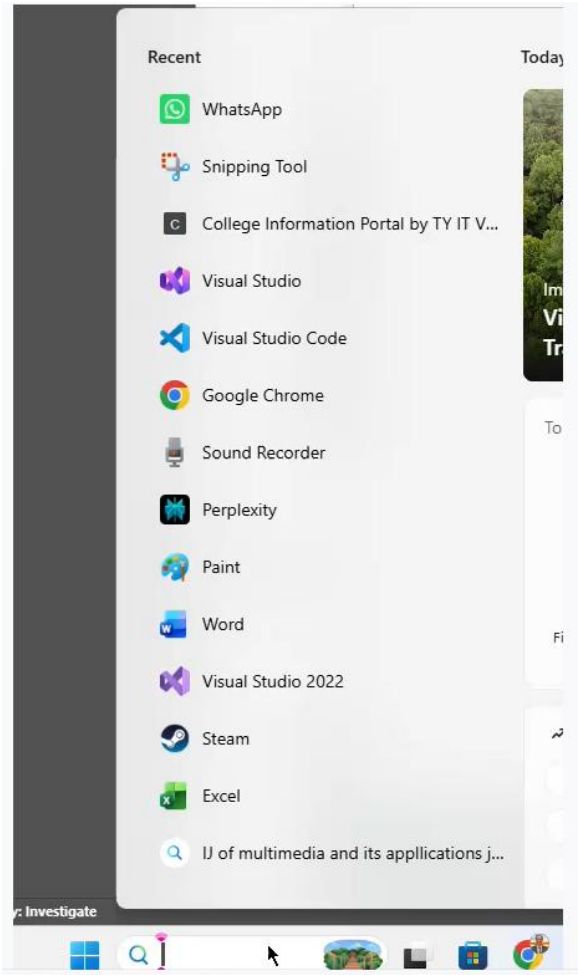
Step 6: Configure Program.cs
Replace the contents of Program.cs with:
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();

app.UseDefaultFiles();
app.UseStaticFiles();

app.Run();
Step 7: Run the PWA
Press **Ctrl + F5**.

Delete manually in Chrome/Edge
Open your website → Press **F12** → **Application** → **Cache Storage** → Find **college-cache** → Right-click → **Delete**.




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